An iterated-log phase transition in rare-event threshold estimation

Specialist review packet

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AI-assisted research draft

28 October 2026

Review brief: a possible iterated-log phase transition

Purpose. This packet asks for a correctness and prior-art assessment of a specific statistical claim. The first three pages give the claim, its vulnerable dependencies, and the verification boundary. The remaining pages contain the complete existing paper argument with its three proof inputs expanded in place. This is an AI-assisted research draft; no external specialist review is claimed.

The experiment and claimed rate

A fresh coin has hidden magnitude D with density $32d$ on $(0, 1/4]$ and hidden sign $S \mid D = d \sim \mathrm{Ber}(q(d))$. Its flips have mean $1/2 + (2S-1)D$. The unknown measurable allocation satisfies $0 \le q \le 1$ and $\theta(q) = \int q\, d\mu \le \rho$, where $0 < \rho \le 1/4$. Algorithms know μ, ρ, B, but observe only their independent randomness and coin flips. They may create, revisit and interleave coins, return biased nonlinear estimates, and stop at an unbounded time. Every flip is charged. The budget is $\sup_q \mathbb{E}_q T \le B$, $B \ge 1$.

For $L = 1 + \log(1/\rho)$ and $H = 1 + \log L$, the paper claims

$$R_{\mu,\rho}(B) \asymp \min\left\{\rho, \sqrt{\frac{\rho L H}{B}}\right\},$$

with universal constants for this one fixed magnitude law. The supremum is over the full global-rarity class; it is not a pointwise constraint $q(d) \le \rho$, and the target is not the known number ρ.

Why this claim matters

The proposed general variational functional has order

$$\Psi_{\mu,\rho}(B^{-1/2}) \asymp \min\left\{\rho, \sqrt{\frac{\rho L}{B}}\right\}$$

at this law. If the new lower bound is correct, no universal matching $C\Psi$ upper bound can hold. For example, the choice $\rho_n = \exp(-\exp n)$, $B_n = \rho_n^{-2}$ gives an unbounded ratio of order at least $\sqrt{H_n}$. The optimizer of the variational functional remains valid; its proposed statistical interpretation is what fails.

The claimed additional cost is the need to discover a coin's sign across unknown gap scales. Sign-entropy mechanisms are established prior work. The proposed contribution is their cost-sensitive transfer to the full population-estimation experiment. No novelty or practical-performance claim follows merely from this explanation.

Proof route and questions for the reviewer

The lower proof uses the convex subclass $\mathcal{X}_\rho = \{0 \le q \le 1/2,\ \int q\, d\mu \le \rho\}$ and $J = \rho LH$. Its logical chain is as follows.

1. **Null-cost entropy obstruction.** For every one-coin estimator with uniform bias at most $\rho/64$ and separately finite uniform variance and work, the claim is

$$\left(\sup_{q \in \mathcal{X}_\rho} \operatorname{Var}_q Y\right) \mathbb{E}_0 T \ge c_b J.$$

   The work is measured under the all-negative population. The proof selects good gaps in separated annuli and restarts the estimator on unused blocks of the *same* coin to construct one sign test. Finite-prefix change-of-measure inequalities and overlapping stopping windows supply the entropy penalty. Fair-coin termination is not assumed.

2. **One common pair.** A finite categorical reduction and convex RMS comparison first give a hard pair for each capped local policy. Revealing the index of a finite policy mixture makes both its null work and squared Hellinger distance exactly affine in the mixture weights. Minimax and weak-star compactness are then used to obtain one separated pair satisfying, for *every* capped policy A,

$$\mathsf{H}^2(P_-^A, P_+^A) \le C\delta^2 \mathbb{E}_0 T_A / J.$$

   The pair is existential. It is not identified with the radial variational optimizer or selected after observing a transcript.

3. **Population transfer and termination.** Finite tree leaves factor by coin even under interleaving. One-coin hybrid simulations use midpoint tapes for the other coins. The midpoint law dominates half the null law, so the one-coin bounds can be charged to actual midpoint work. The same endpoint pair applies at all caps. Finite expected endpoint work then passes the testing bound to the terminal experiment.

Requested scrutiny. Does the formal single-coin model faithfully express the intended *null-cost* theorem? Are the sufficiency, convexity and compactness hypotheses satisfied strongly enough to obtain a pair common to all policies? Does the hybrid simulation charge work under the correct law, including arbitrary randomization and creation? Does the endpoint limit preserve the full unrestricted model?

Separately, is an equivalent critical population-estimation result already known? The complete proof credits Chen–Li–Qiao, Juditsky–Nemirovski, Polyanskiy–Wu and Lee–Valiant. Their ingredients and this particular application should be assessed separately.

What has been checked, and what remains open

Claim or component	Verification boundary
Original arbitrary-law RMS law	Both directions are Lean-checked, with coefficients 1/64 and 512. One budget-indexed upper family works for every gap law in the original observable experiment.
Single-coin base cost	The BC1 variance/null-work bound and its raw-moment version are Lean-checked for all rarities. The proof derives activation, good gaps, observable restarts and the BC2 entropy bound from actual seeded Bernoulli experiments.
Critical minimax lower bound	Lean-checked for the full rare-prior class. The proof constructs one common pair, transfers it to unrestricted population algorithms, and proves endpoint termination and testing. Budget nonemptiness and a trivial upper bound are also proved.
Critical upper and Ψ comparison	Paper proofs with agent review. The matching upper construction and explicit unbounded ratio to Ψ remain unformalized; no external specialist review is claimed.
Numerical and exact finite checks	Sanity and reproducibility evidence. They do not certify the continuum theorem or distinguish its asymptotic rate from the proposed Psi rate.

The existing critical proof was reviewed by an adversarial agent. A second review disclosed that its author also wrote the base-cost component; it must not be counted as a fully independent review of that component. Model diversity is not established. The follow-up audit has not identified a blocking error in its initial pass, but that observation is not a certificate of correctness.

Source and reproduction

The complete mathematical argument below is expanded from the four source files at published commit `b3f88e6bc02b3d59ec9fe99529688e2bd911e1db`. The accompanying build script verifies their existing SHA-256 lock before assembly. Editorial changes in this packet concern the title, this brief, and layout. The inherited mathematical statements and arguments are preserved; monospaced source paths and hashes are made line-breakable for typesetting. The verification table refers to the later formal checkpoint `236eb2d4f286f8987f74f293c129258f517f2ec4`.

- Follow-up dependency audit and formalization boundary.
- Original proof bundle, source lock, and scoped reviews.
- Process note and limits of the evidence.

A matching characterization for arbitrary magnitude laws remains open. This packet concerns a fixed quadratic law. It does not establish uniformity as a power exponent approaches two, explicit useful constants, or practical superiority of an implementation.

Complete critical-law argument

Status. Complete paper proof with an independent adversarial agent review and additional component cross-checks. This is not a Lean proof, an external specialist review, or a novelty certification. Exact reviewed source identities and findings are maintained separately. The strongest claim below concerns the full iid fresh-coin experiment, including nonlinear outputs and interaction among coins; it is not inferred merely from the earlier additive-class theorem.

1 Experiment, quantifiers, and result

Throughout fix $a = 1/4$ and

$$\mu(dd) = \frac{2d}{a^2}\mathbf{1}_{(0,a]}(d)\,dd.$$

For $0 < \rho \le 1/4$, let

$$\mathcal{Q}_\rho = \{q : (0,a] \to [0,1] \text{ measurable} : \int q\,d\mu \le \rho\}, \qquad \theta(q) = \int q\,d\mu.$$

Each genuine fresh coin has an independent hidden magnitude $D \sim \mu$, hidden sign $S \mid D \sim \mathrm{Ber}(q(D))$, and conditionally iid flips of mean $1/2+(2S-1)D$. Neither latent variable is observed. One realization uses independent uniforms on $(0,1)$ for each coin: $D = a\sqrt{U_0}$, $S = \mathbf{1}_{\{U_1 \le q(D)\}}$, and flips $X_k = \mathbf{1}_{\{U_{k+1} \le 1/2+(2S-1)D\}}$ for $k \ge 1$. Measurable functions equal μ-almost everywhere define the same experiment; Borel versions may be used in this standard probability space.

The algorithm has an independent seed and sees only its seed and the observed history. At each step it stops and returns a real measurable output, revisits an observed coin, or observes a fresh coin. Every flip costs one. The output may be biased and nonlinear, and the choices may depend on all previous observations. Stopping may be unbounded. A design may know μ, ρ, B but is fixed before the unknown q is supplied. Work $T = \infty$ on nontermination. Define

$$R_{\mu,\rho}(B) = \inf_{A:\,\sup_{q\in\mathcal{Q}_\rho}\mathbb{E}_q T_A \le B} \sup_{q\in\mathcal{Q}_\rho} \left(\mathbb{E}_q(\widehat{\theta}_A - \theta(q))^2\right)^{1/2}.$$

This class permits arbitrary creation and interleaving, not a globally fixed number of coins. We number coins by their first charged observation. This is equivalent to giving names to iid unseen coins earlier: an unused coin's latent state is independent of the observed history and seed, and choosing among unused names using only that information still selects an iid fresh state. The relabeling retains the map for every later revisit. Creating or discarding an unobserved coin supplies no information. Consequently at most K different coins occur before the Kth charged observation.

Write $L = 1+\log(1/\rho)$, $H = 1+\log L$, and $J = \rho LH$. For the lower proof we use the convex subclass

$$\mathcal{X}_\rho = \{q \in L^\infty(\mu) : 0 \le q \le 1/2 \ \mu\text{-a.e.}, \ \int q\,d\mu \le \rho\}.$$

This pointwise cap restricts the lower-bound adversary only; the algorithmic upper and minimax definition still use the full global-rarity class.

Theorem 1 (Full critical law)**.** *There exist universal constants $0 < c < C < \infty$ such that, for the one fixed law above, every $0 < \rho \le 1/4$ and $B \ge 1$ satisfy*

$$c\min\{\rho, \sqrt{J/B}\} \le R_{\mu,\rho}(B) \le C\min\{\rho, \sqrt{J/B}\}. \tag{A}$$

The lower bound holds for every observable algorithm in the stated full experiment, including arbitrary randomization, nonlinear estimation and data-dependent creation, revisiting and interleaving. The upper permits the known rarity and law; it has separately finite uniform work and squared error.

The upper construction was already available in the separately owned critical note, now at commit `a8a8dc49115aabdd1fba8d53d3390eb0a5eee864`; it is reproduced with attribution in the appendix. The new lower proof has three steps. First, sign-testing entropy forces a small-bias variance product using *all-negative work*, not a supremum of work over populations. Second, convex estimation and minimax supply one hard pair valid for every capped single-coin observation policy. Third, a midpoint simulation charges those bounds to the actual work of an interactive population algorithm. The last step is the established Lee–Valiant Hellinger decomposition mechanism [3, Lemma 17], proved below at finite prefixes to make its observation and stopping scope explicit.

The convex-estimation step follows Juditsky–Nemirovski's saddle-point framework [1]; the passage from its exponential tails to RMS risk is written out here. Polyanskiy–Wu [2] already establish broad quadratic-risk affine comparisons and convex duality for functional estimation. Neither that general principle nor the Hellinger decomposition is claimed as new. The cost-sensitive common pair and its application to this critical hidden-coin class are the proposed contribution; priority remains unverified.

2 A single-coin entropy bound with all-negative reference cost

Fix $a = 1/4$ and $\mu(dd) = 2da^{-2}\mathbf{1}_{(0,a]}(d)\,dd$. Throughout this section,

$$\mathcal{X}_\rho = \left\{ q : 0 \le q \le \tfrac{1}{2},\ \int q\,d\mu \le \rho \right\}, \qquad 0 < \rho \le \tfrac{1}{4}, \qquad L = 1 + \log(1/\rho), \quad H = 1 + \log L.$$

All functions defining populations are measurable. A single genuine coin has hidden $D \sim \mu$ and $S \mid D = d \sim \mathrm{Ber}(q(d))$; conditionally on (D, S) its observations are iid Bernoulli with parameter $1/2 + (2S-1)D$. A policy receives only its independent seed and observed bits. It may know μ, a, ρ, but not q, D, S.

Theorem 2 (The critical product with null-population cost)**.** *There is a universal $c_0 > 0$ such that the following holds for every $0 < \rho \le 1/4$. Let an observable single-coin procedure have real output Y and stopping time $T \in \mathbb{N}_0 \cup \{\infty\}$, and suppose*

$$\sup_{q\in\mathcal{X}_\rho} |\mathbb{E}_q Y - \theta(q)| \le \rho/64, \qquad V := \sup_{q\in\mathcal{X}_\rho} \mathrm{Var}_q\, Y < \infty, \qquad \sup_{q\in\mathcal{X}_\rho} \mathbb{E}_q T < \infty, \qquad \theta(q) = \int q\,d\mu.$$

Writing $W_0 = \mathbb{E}_{q=0} T$, one has

$$VW_0 \ge c_0 \rho L H. \tag{BC1}$$

The same order lower bound holds with $M = \sup_{q\in\mathcal{X}_\rho} \mathbb{E}_q Y^2$ in place of V. The cost is the expectation under the all-negative population, rather than the supremum of expected work over the parameter class. The policy may stop before observing the coin and may have unbounded stopping.

The proof adapts the separately documented critical-two single-coin argument. We reproduce its finite-gap dependency to make this statement self-contained. Its entropy mechanism follows Chen–Li–Qiao, arXiv:1608.06031, Lemma 4.1; the Bernoulli, one-sided-cost statement used here is proved explicitly below. No novelty is claimed for that mechanism.

Lemma 3 (Finite-gap sign entropy)**.** *There are universal $c_s > 0$ and $K_s < \infty$ with the following property. Let $d_1 > \cdots > d_K > 0$ satisfy $d_j/d_{j+1} \ge 2$ and $d_j \le 1/128$. Suppose one observable sign-testing procedure, the same at all these gaps, has error at most $1/10$ under every Bernoulli law $1/2 \pm d_j$, and finite expected stopping time τ under both signs. For $K \ge K_s$,*

$$\sum_{j=1}^{K} d_j^2 \mathbb{E}_{-d_j} \tau \ge c_s K \log K. \tag{BC2}$$

Correctness and termination under the fair coin are not required. One may take $c_s = 1/512$ *and* $K_s = \lceil e^{32} \rceil$.

Proof. Put $b_0 = 1/1024$, $n_j = \lfloor b_0/d_j^2 \rfloor$, $u_j = \lceil 4\mathbb{E}_{-d_j}\tau \rceil$, $c_j = d_j^2 \mathbb{E}_{-d_j}\tau$, and $\bar{c} = K^{-1}\sum_j c_j$. Since $d_j \le 1/128$, $n_j \ge b_0/(2d_j^2) \ge 1$. The event of returning minus by time n_j is measurable in the independent seed and the first n_j flips, including unused coordinates if the test has stopped. The KL divergence between these full prefixes at the minus and plus laws is at most

$$n_j \frac{(2d_j)^2}{(1/2+d_j)(1/2-d_j)} \le \frac{64}{3} n_j d_j^2 \le \frac{1}{48}.$$

Bernoulli KL is at most its chi-square divergence. Pinsker's inequality bounds the total variation by less than $1/8$. Since an early minus is an error under the plus law, it has probability at most $1/10 + 1/8$ under the minus law. Markov's inequality gives $\mathbb{P}_{-d_j}\{\tau > u_j\} \le 1/4$. Consequently

$$A_j = \{\text{output minus and } n_j < \tau \le u_j\} \quad \text{satisfies} \quad \mathbb{P}_{-d_j}(A_j) \ge 1 - \tfrac{1}{10} - (\tfrac{1}{10} + \tfrac{1}{8}) - \tfrac{1}{4} > \tfrac{1}{4}.$$

Let P_0 denote the fair-coin law with the same seed and policy. Comparing the deterministic prefixes of length u_j, their KL divergence from minus to fair is at most $4d_j^2 u_j \le 16c_j + 1$. For an event of probability $p \ge 1/4$ under the minus law and probability z under the fair law, binary data processing and $\mathrm{kl}(p,z) \ge p\log(1/z) - \log 2$ therefore give

$$P_0(A_j) \ge e^{-64c_j - 8}. \tag{BC3}$$

Only deterministic finite prefixes have been compared; this does not invoke optional stopping under a possibly nonterminating fair law.

At least $K/2$ indices form a set G with $c_j \le 2\bar{c}$. For them,

$$(n_j, u_j] \subset [b_0/(2d_j^2), (8\bar{c}+1)/d_j^2].$$

The scales d_j^{-2} have successive ratios at least four. At any finite time the number of these intervals containing that time is at most $1 + \log_4((16\bar{c}+2)/b_0)$. All events A_j refer to the same terminal time and output on a realization under the fair law; on a nonterminating path none occurs. Summing their indicators and using (BC3) yields

$$(K/2)e^{-128\bar{c}-8} \le 1 + \log_4(1024(16\bar{c}+2)). \tag{BC4}$$

If $x = \log K \ge 32$ and $\bar{c} < x/512$, the left side exceeds $\frac{1}{2}e^{3x/4-8} \ge \frac{1}{2}e^{x/2} \ge x^2/16$. The right side is less than $1 + \log_4(32x + 2048) < 6 + x/4$, contradicting $x \ge 32$. Thus $\bar{c} \ge (\log K)/512$, as claimed. □

Proof of Theorem 2. The bias bound implies $M \le V + (65\rho/64)^2 < \infty$. First define the nonnegative extended conditional moments

$$M_s(d) = \mathbb{E}_{d,s} Y^2, \qquad w_s(d) = \mathbb{E}_{d,s} T.$$

The admissible populations $q = 0$ and $q \equiv \rho$ imply $\int M_- d\mu \le M$, $\int M_+ d\mu \le M/\rho$, and finite integrals of both work functions. All conditional quantities used below are thus finite outside a common null set. Define signed conditional means there, and give them arbitrary values on that null set. For $f(d) = \mathbb{E}_{d,+}Y - \mathbb{E}_{d,-}Y$, the two bias bounds imply

$$\int f \, d\mu = (\mathbb{E}_\rho Y - \mathbb{E}_0 Y)/\rho \ge 31/32.$$

Conditioning on the latent magnitude and then on its sign at the constant-prior population gives

$$V \ge \mathrm{Var}_\rho(Y) \ge \rho(1-\rho) \int f^2 \, d\mu \ge \rho(1-\rho)(31/32)^2 \ge 2\rho/3. \tag{BC5}$$

This uses only the bias guarantee, not a pointwise unbiased response. It also shows

$$M \le \tfrac{3}{2} V. \tag{BC6}$$

Normalize the first observation. The event $A = \{T \geq 1\}$ is decided by the independent seed before any observation, so its probability p is independent of the latent state. It is positive: otherwise the expectations at target values zero and ρ could not both satisfy the bias bound. The finite constant $c = \mathbb{E}[Y\mathbf{1}_{A^c}]$ is also population independent. Under the seed law conditioned on A, form the active procedure

$$Z = pY + c, \qquad T_Z = T.$$

It preserves every expectation of Y and always takes at least one observation. The conditioning is implementable by resampling independent seeds before observing the coin; its constants depend on the specified policy and known laws, not on q, D, S.

For $p < 1$, expansion, Cauchy–Schwarz on A^c, and completing the square give

$$\begin{aligned}\mathbb{E}_q Z^2 &= p\mathbb{E}_q[Y^2\mathbf{1}_A] + 2c\mathbb{E}_q Y - c^2 \\ &\leq pM + 2c\mathbb{E}_q Y - c^2/(1-p) \\ &\leq pM + (1-p)(\mathbb{E}_q Y)^2 \leq m := pM + 2\rho^2.\end{aligned}$$

For $p = 1$, take $c = 0$ and the final bound holds directly. Apply (BC5) to Z to obtain $m \geq 2\rho/3$ and $pM \geq 2\rho/3 - 2\rho^2 \geq \rho/6$. Set $P = MW_0$ and $W_{0Z} = \mathbb{E}_0 T_Z = W_0/p \geq 1$. Then

$$m \leq 4pM \leq 4P, \qquad mW_{0Z} \leq 4P, \qquad P \geq \rho/6. \tag{BC7}$$

Indeed $2\rho^2 \leq \rho/2 \leq 3pM$. Every work quantity in (BC7) is all-negative work.

Select good gaps within the capped global class. Now use conditional means and moments for Z, keeping the same notation. The bad set $B_- := \{M_- > 1/64\}$ has mass at most $64m$. The set

$$B_+ := \{M_+ - M_- > 2m/\rho\}$$

has mass less than 2ρ. Otherwise nonatomicity supplies $E \subset B_+$ of mass 2ρ, and the admissible $q = \frac{1}{2}\mathbf{1}_E$ gives

$$\mathbb{E}_q Z^2 = \int M_- d\mu + \tfrac{1}{2}\int_E (M_+ - M_-)d\mu > 2m,$$

contradicting its bound. The response bad set $B_f := \{f < 3/4\}$ also has mass less than 2ρ. If not, choose such an E of mass 2ρ. Its population has target ρ, but

$$\mathbb{E}_{\mathbf{1}_E/2} Z - \mathbb{E}_0 Z = \tfrac{1}{2}\int_E f\, d\mu < 3\rho/4,$$

whereas the bias bounds force at least $31\rho/32$.

Remove these bad sets and the conditional integrability null sets. The remaining good set G has $\mu(G^c) \leq 64m + 4\rho \leq 70m$. On it,

$$|\mathbb{E}_- Z| \leq 1/8, \qquad \mathbb{E}_+ Z \geq 5/8, \qquad M_- \leq 1/64, \qquad M_+ \leq 1/64 + 2m/\rho. \tag{BC8}$$

The use of sets of mass 2ρ is necessary for the cap $q \leq 1/2$; the argument does not substitute a pointwise bound $q \leq \rho$.

Build a common sign test. Let $n = \lceil 512(1 + m/\rho)\rceil \leq 1282m/\rho$, the inequality using $m/\rho \geq 2/3$. On the same fixed-gap coin, restart the active procedure n times with independent seeds and consecutive unused flip blocks. Output plus if their mean exceeds $3/8$. At a fixed latent state the restarts are iid: condition on each finite stopping prefix, use the independence of its unused iid suffix, and sum over the countably many prefixes. Finite conditional expected work ensures almost-sure termination for both signs at each good gap. No new latent magnitude

is drawn. The same test is used at every good gap. By (BC8), either mean is at least $1/4$ from the threshold, and Chebyshev bounds either error by

$$\frac{16(1/64 + 2m/\rho)}{n} < 1/10.$$

Its negative expected work at gap d is exactly $nw_-(d)$.

Average negative work over separated annuli. For $j \geq 3$, put

$$I_j = (a4^{-j}/2, a4^{-j}], \qquad \mu(I_j) = \tfrac{3}{4}16^{-j}, \qquad J = \left\lfloor \log_{16} \frac{3}{560m} \right\rfloor, \qquad K = J - 2.$$

When $J \geq 3$, every retained annulus has mass at least $140m$, so G occupies at least half of it. Choose $d_j \in G \cap I_j$ with

$$w_-(d_j) \leq \frac{4}{\mu(I_j)} \int_{I_j} w_- \, d\mu.$$

This follows by averaging; the integral is positive since the active procedure always observes at least once. The choices are existential proof objects, not inputs to the policy. Successive gap ratios exceed two, every gap is at most $1/256$, and

$$\sum_{j=3}^{J} d_j^2 w_-(d_j) \leq \frac{16a^2}{3} \int w_- d\mu = \frac{16a^2}{3} W_{0Z}.$$

Lemma 3, the restart bound, and (BC7) imply, for $K \geq K_s$,

$$c_s K \log K \leq \sum_{j=3}^{J} d_j^2 n w_-(d_j) \leq \frac{16a^2}{3} \frac{1282m}{\rho} W_{0Z} \leq \frac{30000a^2 P}{\rho}. \tag{BC9}$$

The last numerical constant is conservative, since $(16/3) \cdot 1282 \cdot 4 < 30000$.

Obtain one positive constant for every rarity. For sufficiently small ρ, if $P \geq \sqrt{\rho}$ the desired order already follows because $\sqrt{\rho} LH \to 0$. Otherwise (BC7) gives $m \leq 4\sqrt{\rho}$ and

$$K \geq \frac{\frac{1}{2}\log(1/\rho) - \log(2240/3)}{\log 16} - 3.$$

For all sufficiently small universal ρ this exceeds K_s, is at least a positive universal multiple of L, and satisfies $\log K \geq \frac{1}{2}\log L$. Equation (BC9), with the fixed $a = 1/4$, gives $P \geq c\rho LH$. On the remaining compact interval of rarities, the lower bound $P \geq \rho/6$ in (BC7) allows the same conclusion after decreasing one positive universal constant. Finally (BC6) transfers the raw-moment lower bound to VW_0, proving (BC1). □

The all-negative cost is the strengthened part of this statement. It does not interchange separate suprema, assert a same-population product, or by itself reduce an arbitrary population algorithm to a single coin.

3 An RMS comparison for finite convex experiments

Let X be a finite-dimensional nonempty compact convex set and $\ell(x)$ a linear functional. Consider either of the following experiments:

1. independent finite categorical observations, whose positive probability vectors $p_i(x)$ are affine functions of x;

2. a finite vector of independent Poisson counts, whose strictly positive intensities $v_j(x)$ are affine functions of x.

In the first case let $\mathcal{F}$ be the finite-dimensional space of functions $c + \sum_i \phi_i(z_i)$. In the second let $\mathcal{F} = \{c + \sum_j \beta_j z_j\}$. Let P_x denote the whole observation law. All exponential moments of functions in $\mathcal{F}$ are finite. For fixed $\phi \in \mathcal{F}$, the function $x \mapsto \log \mathbb{E}_x e^{\phi}$ is concave: it is a sum of logarithms of positive affine forms in the categorical case, and an affine function of the intensities in the Poisson case. Moreover $\log(dP_x/dP_y) \in \mathcal{F}$.

Lemma 4 (RMS affine comparison)**.** *For any measurable estimator G with $r_G = \sup_{x\in X}(\mathbb{E}_x(G-\ell(x))^2)^{1/2} < \infty$ and any $\eta > 0$, there exists $F \in \mathcal{F}$ such that*

$$\sup_{x\in X}(\mathbb{E}_x(F-\ell(x))^2)^{1/2} \le 12 r_G + \eta. \tag{2}$$

The result permits nonattainment of the infimum and imposes no restriction on the original estimator's output range.

Proof. Put $s = \log 2$. For $\alpha > 0$ and $\phi \in \mathcal{F}$ define

$$\Phi(x, y; \phi, \alpha) = \ell(x) - \ell(y) + \alpha \log \mathbb{E}_y e^{\phi/\alpha} + \alpha \log \mathbb{E}_x e^{-\phi/\alpha} + 2\alpha s,$$
$$2S = \inf_{\phi,\alpha>0} \max_{x,y\in X} \Phi(x, y; \phi, \alpha).$$

This real-valued continuous function is convex in (ϕ, α) by the perspective transformation and concave in (x, y) by the preceding log-moment observation. Sion's minimax theorem applies because $X \times X$ is compact convex and the other domain is convex. Thus the infimum and supremum commute. For fixed x, y, Cauchy–Schwarz gives

$$\inf_{\psi\in\mathcal{F}} \left(\log \mathbb{E}_y e^{\psi} + \log \mathbb{E}_x e^{-\psi}\right) = 2 \log \mathrm{Aff}(P_x, P_y),$$

with equality at $\psi = \frac{1}{2}\log(dP_x/dP_y)$. Taking the infimum over $\alpha > 0$ after putting $\phi = \alpha\psi$ therefore proves

$$2S = \max\{\ell(x) - \ell(y) : x, y \in X,\ \mathrm{Aff}(P_x, P_y) \ge 1/2\}. \tag{3}$$

The constraint is closed and nonempty, so the maximum exists. Continuity of affinity is immediate from the finite categorical formula and, for Poisson vectors, from $\mathrm{Aff}(P_x, P_y) = \exp[-\frac{1}{2}\sum_j(\sqrt{v_j(x)} - \sqrt{v_j(y)})^2]$. In particular $0 \le S < \infty$.

For the maximizing pair in (3), total variation is at most $\sqrt{3}/2$ by the standard Cauchy–Schwarz affinity bound $\|P - Q\|_{\mathrm{TV}} \le \sqrt{1 - \mathrm{Aff}(P, Q)^2}$. Testing the two endpoints by thresholding G at their functional midpoint gives, with $\Delta = 2S$,

$$2r_G^2 \ge \frac{\Delta^2}{4}(1 - \|P_x - P_y\|_{\mathrm{TV}}), \qquad S \le \sqrt{\frac{2}{1 - \sqrt{3}/2}}\, r_G. \tag{4}$$

Indeed each wrong decision has squared estimation error at least $\Delta^2/4$, and the sum of the two testing errors is at least one minus total variation. The case $S = 0$ requires no division by S.

Fix $R > S$. Choose (ϕ, α) with $\max_{x,y} \Phi \le 2R$. On the diagonal, Cauchy–Schwarz gives $\Phi(x, x; \phi, \alpha) \ge 2\alpha s$, hence $\alpha \le R/s$. Set

$$U = \max_x\{\ell(x) + \alpha \log \mathbb{E}_x e^{-\phi/\alpha}\}, \qquad V = \max_x\{-\ell(x) + \alpha \log \mathbb{E}_x e^{\phi/\alpha}\}.$$

Then $U + V + 2\alpha s \le 2R$. The additive estimator $F = \phi + (U - V)/2$ satisfies, uniformly in x,

$$\mathbb{E}_x e^{(F-\ell(x))/\alpha} \le e^{R/\alpha - s}, \qquad \mathbb{E}_x e^{(\ell(x)-F)/\alpha} \le e^{R/\alpha - s}.$$

For $t \geq 0$, Markov's inequality consequently gives $\mathbb{P}_x(|F - \ell(x)| > R + t) \leq e^{-t/\alpha}$, since $2e^{-s} = 1$. Integrating this tail, rather than substituting a confidence guarantee for an RMS bound, yields

$$\mathbb{E}_x(F - \ell(x))^2 \leq R^2 + 2R\alpha + 2\alpha^2 \leq R^2 \left(1 + \frac{2}{s} + \frac{2}{s^2}\right). \tag{5}$$

The product of the squared constants in (4)–(5) is less than 144: $\log 2 > 2/3$ and $\sqrt{3}/2 < 7/8$ bound it by $8.5 \cdot 16 = 136$. Letting $R \downarrow S$, with the slack chosen for the given η, proves (2). This also proves uniform square integrability of the produced estimator, rather than assuming it. □

4 From a base-cost obstruction to one pair hard for every capped policy

In this section only, restrict the parameter set to

$$\mathcal{X}_\rho = \{q : 0 \leq q \leq 1/2, \int q \, d\mu \leq \rho\}, \qquad J = \rho LH.$$

It is a weak-star compact convex subset of $L^\infty(\mu)$. The following implication separates its one required entropy input from the convex-experiment argument. The input is the base-cost bound proved in the accompanying base-cost section of the full-model note:

There is a universal $c_b > 0$ such that every observable single-coin estimator Y with
$\sup_{q \in \mathcal{X}_\rho} |\mathbb{E}_q Y - \theta(q)| \leq \rho/64$, $V = \sup_{q \in \mathcal{X}_\rho} \mathrm{Var}_q Y < \infty$, $\sup_{q \in \mathcal{X}_\rho} \mathbb{E}_q T < \infty$ satisfies $V \mathbb{E}_0 T \geq c_b J$. (BC)

The cost in (BC) is at the all-negative population $q = 0$, not the supremum over populations. The two lemmas below are conditional on precisely this input. No algorithm-level lower bound is inferred from a variational optimization alone. Write $h(P, Q) = 1 - \mathrm{Aff}(P, Q)$ for squared Hellinger distance in the normalization used here; it takes values in $[0, 1]$.

Exact finite observation reduction

A capped local policy A includes its independent seed and stops after at most $m < \infty$ flips of one genuine coin. Its full observed transcript contains the seed and the stopped word. Its seed distribution and policy may depend on known parameters, but not on the actual q. For each word $x \in \{0, 1\}^t$, let $B_{t,x}$ be the measurable seed set that causes stopping exactly at that word. For $0 \leq k \leq t \leq m$ put

$$a_{t,k} = \sum_{x:\, |x| = t,\, \#1(x) = k} \mathbb{P}(\text{seed} \in B_{t,x}),$$

and, with $p_s(d) = 1/2 + (2s - 1)d$, put

$$m_{t,k}(q) = \int \left[(1 - q(d))p_0(d)^k (1 - p_0(d))^{t-k} + q(d)p_1(d)^k (1 - p_1(d))^{t-k}\right] d\mu.$$

The finite statistic (T, K), where K is the number of observed heads, has probabilities $a_{t,k} m_{t,k}(q)$. Conditional on (T, K), the stopped word and seed have a distribution independent of q: their common likelihood multiplier is exactly $m_{t,k}(q)$ and cancels. This statement also follows directly by integrating any bounded transcript function over the finitely many word/seed cells, so it does not assume a latent oracle. Consequently the full transcript and this finite statistic have exactly the same Hellinger affinities for all pairs of populations.

Remove coordinates with $a_{t,k} = 0$. The remaining coordinates are positive for every q, since $p_s(d) \in [1/4, 3/4]$. They are affine weak-star continuous functions of q. Thus their probability

vector together with $\theta(q)$ has a compact convex finite-dimensional range. In particular, for each capped A, its full-transcript squared Hellinger distance $h_A(q_-, q_+)$ is a continuous convex function of the pair on $\mathcal{X}_\rho^2$. Convexity follows from joint concavity of affinity for two affine probability vectors. This reduction retains the stopping time and expected cost, not just a fixed number of sampled bits.

Lemma 5 (A hard pair for each capped policy)**.** *Assume (BC). There are universal constants $c_\delta > 0$ and $C_H < \infty$ such that, for every $\rho \in (0, 1/4]$, every $0 < \delta \le c_\delta \rho$, and every randomized capped local policy A, there exist $q_-, q_+ \in \mathcal{X}_\rho$ with $\theta(q_+) - \theta(q_-) \ge \delta$ and*

$$h_A(q_-, q_+) \le C_H \frac{\delta^2}{J} \mathbb{E}_0 T_A. \tag{6}$$

The full observed seed and stopped history are included in the Hellinger distance.

Proof. Let $w = \mathbb{E}_0 T_A$. If $w = 0$, the seed-only event of taking a first observation has probability zero, so A has the same transcript law for every q. The pair $q_- = 0$, $q_+ = \delta$ gives separation δ and $h_A = 0$. We therefore assume $w > 0$ in all the divisions below. First establish the following additive lower bound for $n \ge 1$ independent runs of A. If an additive output with bounded component functions $F = c + \sum_{i=1}^n Y_i$ has worst RMS $r < \rho/64$, put $b_i = \mathbb{E}_0 Y_i$, $b = \mathbb{E}_0 F$, and choose I uniformly in $\{1, \dots, n\}$ independently. Run just its component on one coin and return $Z = n(Y_I - b_I) + b$. This preserves every population mean and uses base cost w.

For $\theta = \theta(q)$, the laws of each component output under q and 0 have a common subprobability of mass $1 - \theta$: integrate the negative-sign kernel with weight $1 - q(d)$. For any two distributions sharing such a subprobability, writing each as a mixture of its normalized common part and residual part shows

$$(\mathbb{E}_q Y_i - \mathbb{E}_0 Y_i)^2 \le \frac{2\theta}{1-\theta} (\operatorname{Var}_q Y_i + \operatorname{Var}_0 Y_i). \tag{7}$$

Indeed the squared displacement of either mixture mean from the common mean is at most $\theta/(1-\theta)$ times its variance; use $(u - v)^2 \le 2u^2 + 2v^2$. The case $\theta = 0$ is identical laws. Independence and $\sup_q \operatorname{Var}_q F \le r^2$ imply

$$\sum_i (\mathbb{E}_q Y_i - b_i)^2 \le \frac{4\rho}{1-\rho} r^2, \quad \operatorname{Var}_q Z \le n \sum_i \operatorname{Var}_q Y_i + n \sum_i (\mathbb{E}_q Y_i - b_i)^2 \le \frac{7}{3} n r^2.$$

All outputs here are bounded, and stopping is capped, so (BC) applies to Z without an additional integrability assumption. It gives $r^2 \ge 3c_b J/(7nw)$. Including the branch $r \ge \rho/64$ establishes

$$r \ge c_a \min\{\rho, \sqrt{J/(nw)}\}, \qquad c_a = \min\{1/64, \sqrt{3c_b/7}\}. \tag{8}$$

For $w > 0$, let S_n be the saddle quantity S from the proof of Lemma 4 for n independent copies of the finite statistic of A, on the parameter set $\mathcal{X}_\rho$. Equations (3) and (5) of that proof show

$$2S_n = \max\{\theta(q_+) - \theta(q_-) : \operatorname{Aff}(P_{A,q_+}, P_{A,q_-})^n \ge 1/2\}, \quad \inf_{F \text{ additive}} \sup_q \|F - \theta(q)\|_2 \le 3S_n.$$

The latter constant is valid because $1 + 2/\log 2 + 2/(\log 2)^2 < 9$. The finite statistic outputs are bounded, so (8) applies. Hence, for $\beta = 2c_a/3$,

$$2S_n \ge \beta \min\{\rho, \sqrt{J/(nw)}\}. \tag{9}$$

Take $c_\delta = \beta/2$ and $C_H = 8/\beta^2$. For $0 < \delta \le c_\delta \rho$ set $x = \beta^2 J/(4\delta^2 w)$. If $x \ge 2$, choose $n = \lfloor x \rfloor$, so $x/2 \le n \le x$. Equation (9) gives a pair of separation at least 2δ, with

$$h_A \le 1 - e^{-(\log 2)/n} \le \frac{\log 2}{n} \le \frac{8 \log 2}{\beta^2} \frac{\delta^2 w}{J} \le C_H \frac{\delta^2 w}{J}.$$

If $x < 2$, the pair $q_- = 0$, $q_+ = \delta$ belongs to $\mathcal{X}_\rho$, has separation δ, and $h_A \le 1$ suffices because $C_H \delta^2 w/J > 1$. □

Lemma 6 (One pair hard for every capped policy)**.** *Assume (BC), and keep the universal constants of Lemma 5. For every $0 < \rho \le 1/4$ and $0 < \delta \le c_\delta \rho$, there exist $q_-, q_+ \in \mathcal{X}_\rho$, with $\theta(q_+) - \theta(q_-) \ge \delta$, such that simultaneously for every randomized capped single-coin policy A,*

$$h_A(q_-, q_+) \le C_H \frac{\delta^2}{\rho LH} \mathbb{E}_0 T_A. \tag{10}$$

Each policy's seed and full stopped observable history are included. The constants and pair are chosen before the policy.

Proof. Let $\mathcal{K}_\delta = \{(q_-, q_+) \in \mathcal{X}_\rho^2 : \theta(q_+) - \theta(q_-) \ge \delta\}$. It is nonempty compact convex in the weak-star topology. Fix any finite list of capped policies $A_1, \dots, A_k$ and put $w_i = \mathbb{E}_0 T_{A_i}$ and $\lambda = C_H \delta^2/(\rho LH)$. For a simplex vector $\pi \in \Delta_k$, define the observable mixture policy that draws $i \sim \pi$, reveals this index in its independent seed, and runs A_i. This is still capped, by the largest of the finitely many caps. Its base cost is $\sum_i \pi_i w_i$ and, since the revealed labels occupy disjoint transcript components, its squared Hellinger distance is exactly $\sum_i \pi_i h_{A_i}$. Lemma 5 therefore gives

$$\min_{z \in \mathcal{K}_\delta} \sum_i \pi_i \big(h_{A_i}(z) - \lambda w_i\big) \le 0 \qquad (\pi \in \Delta_k).$$

For the finite simplex the displayed function is continuous affine in π and continuous convex in z, by the finite observation reduction. Sion's theorem now gives

$$\min_{z \in \mathcal{K}_\delta} \max_{1 \le i \le k} \big(h_{A_i}(z) - \lambda w_i\big) = \max_{\pi \in \Delta_k} \min_{z \in \mathcal{K}_\delta} \sum_i \pi_i \big(h_{A_i}(z) - \lambda w_i\big) \le 0.$$

Thus the closed subsets $\{z \in \mathcal{K}_\delta : h_A(z) \le \lambda \mathbb{E}_0 T_A\}$ have the finite intersection property as A ranges over all capped policies. Compactness of $\mathcal{K}_\delta$ gives a point in their full intersection, proving (10). This uses only finite revealed mixtures and a finite simplex minimax theorem; no topology on a space of all policies, unjustified exchange of arbitrary infima, or unobserved-mixture affinity identity is needed. □

5 From one universal pair to every population algorithm

Lemma 7 (Finite-prefix midpoint decomposition)**.** *Suppose $q_-, q_+ \in \mathcal{X}_\rho$ have this property: every observable randomized single-coin policy with a deterministic finite cap obeys*

$$\mathsf{H}^2(P^A_{q_-}, P^A_{q_+}) \le \lambda \mathbb{E}_0 T_A, \tag{B}$$

where the transcript includes its independent seed and stopped observations, and $\mathsf{H}^2(P, Q) = 1 - \int \sqrt{dP\, dQ}$. Let $\bar{q} = (q_- + q_+)/2$. Then every population algorithm capped after K total flips obeys

$$\mathsf{H}^2(P^{\mathrm{pop}}_{q_-}, P^{\mathrm{pop}}_{q_+}) \le 2\lambda \mathbb{E}_{\bar{q}} T. \tag{C}$$

Proof. Condition first on the original independent seed. Pad the list of observed coins to K slots by their first-observation indices; an unobserved slot contributes the empty word of probability one. The policy is a finite binary decision tree, whose choices are determined by past outcomes and the fixed seed. For a leaf z, let $h_i(z)$ be the ordered word observed from slot i. Under a single fresh coin from q_r, write its word probability as $p_r(h)$, $r \in \{-, +\}$. Independence of the coin tapes gives leaf density

$$P_r(z) = \prod_{i=1}^{K} p_r(h_i(z)).$$

Interleaving changes the tree and the words, but not this factorization. Set $m_i = (p_-(h_i) + p_+(h_i))/2$ and $c_i = \sqrt{p_-(h_i) p_+(h_i)}$. Word probabilities are affine in q, so $m_i = p_{\bar{q}}(h_i)$. These

quantities are positive; empty words cause no exception. Since $0 \le c_i \le m_i$, the elementary product inequality gives

$$\prod_i m_i - \prod_i c_i \le \sum_i (m_i - c_i) \prod_{j \ne i} m_j. \tag{D}$$

The left side sums over leaves to $\mathsf{H}^2(P_-^{\mathrm{pop}}, P_+^{\mathrm{pop}})$, because the midpoint experiment is normalized. The ith right side sums to the squared Hellinger distance between the two hybrid population experiments having slot i drawn from q_- or q_+ and all other slots independently drawn from $\bar{q}$. Those finite experiments are normalized too. Integrating the original common seed preserves this inequality: joint Hellinger distance with a revealed common seed is the integral of conditional Hellinger distances.

For each slot i, realize the hybrid experiment by a single-coin policy A_i. Simulate all other slots using independent artificial tapes from the fixed known reference population $\bar{q}$, and supply slot i by one genuine coin. If that slot is never requested, take no genuine observations. Include the original seed and all artificial tapes in the independent auxiliary seed. This is an observable single-coin policy capped at K genuine flips. It may use the fixed reference pair as design constants, but these do not depend on the genuine coin's unknown endpoint law. The universal quantifier in (B) covers this policy. No artificial tape is supplied to the original population algorithm.

The hybrid population transcript is a measurable function of this larger seed and the stopped genuine observations. Hellinger data processing and (B) therefore bound its distance by $\lambda \mathbb{E}_0 T_{A_i}$. Since $0 \le \bar{q} \le 1/2$, the one-coin latent distribution under $\bar{q}$ dominates one half of the all-negative distribution, with the magnitude law fixed. The same domination holds after adjoining the common auxiliary seed and taking any measurable nonnegative work functional. Hence

$$\mathbb{E}_0 T_{A_i} \le 2 \mathbb{E}_{\bar{q}} T_{A_i}.$$

When the genuine coin too is from $\bar{q}$, its expected charged work is exactly the work on slot i in the original midpoint experiment. Summing over the at most K slots yields (C). All simulations are only devices in the lower-bound comparison; every genuine flip of the population algorithm is counted exactly once. □

Proof of the lower bound in Theorem 1. Take the universal constants c_δ, C_H from Lemma 6. Choose a fixed

$$0 < \gamma \le \min\{c_\delta, 1/(4\sqrt{C_H})\}, \qquad \delta = \gamma \min\{\rho, \sqrt{J/B}\}.$$

That lemma provides one pair $q_-, q_+ \in \mathcal{X}_\rho$ with target separation at least δ and property (B) with $\lambda = C_H \delta^2 / J$. The pair is valid simultaneously for all finite caps and policies; it is not reselected after seeing a population transcript.

Let A be any full population algorithm with uniform expected work at most B. Apply Lemma 7 to its prefix capped after K flips, recording whether it already halted and, when relevant, its terminal output as a measurable statistic. Since $\bar{q} \in \mathcal{X}_\rho \subset \mathcal{Q}_\rho$, its midpoint expected capped work is at most B. Thus every prefix has squared Hellinger distance at most $2C_H \delta^2 B / J \le 1/8$. The inequality $\mathrm{TV}(P, Q) \le \sqrt{2\mathsf{H}^2(P, Q)}$ bounds its total variation by $1/2$.

Uniform expected work implies almost-sure termination separately under both endpoints. For any terminal event E, the event $E \cap \{T \le K\}$ is determined by the capped prefix, and the omitted endpoint probabilities are at most $\mathbb{P}_{q_-}(T > K) + \mathbb{P}_{q_+}(T > K) \to 0$. Consequently the terminal transcript's total variation is also at most $1/2$. This passage assumes neither a globally bounded number of coins nor differentiability or a chain rule at an unbounded stopping time.

If the estimator has infinite worst-case squared error, the lower bound is automatic. Otherwise threshold its output halfway between the endpoint targets. With endpoint separation $\Delta \ge \delta$ and worst RMS r, Markov's inequality bounds the sum of the two testing errors by $8r^2/\Delta^2$. Every test has error sum at least $1 - \mathrm{TV} \ge 1/2$. Hence $r \ge \Delta/4 \ge \delta/4$, which proves the lower inequality in (A), including saturation. □

6 Consequence for the proposed rare-event modulus

For this exact law the variational optimizer gives a cutoff $t \in (0, a)$:

$$\rho = (t/a)^2[1 + 2\log(a/t)], \qquad \Psi_{\mu,\rho}(h) = \begin{cases} [2ht(1 + \log(a/t)) - h^2]/a^2, & 0 < h < t, \\ \rho, & h \geq t. \end{cases} \tag{E}$$

These are inherited variational identities, not an attaining statistical algorithm. Put $u = 1 + 2\log(a/t)$. Then $\rho = ue^{1-u}$ and $L = u - \log u$, so $L \leq u \leq 2L$. For $h < t$, $\Psi(h) = (ht/a^2)(u + 1 - h/t)$; its coefficient lies between u and $u+1$. Together with the saturated branch this proves, with universal constants and fixed $a = 1/4$,

$$\Psi_{\mu,\rho}(B^{-1/2}) \asymp \min\{\rho, \sqrt{\rho L/B}\}.$$

For instance take $\rho_n = \exp(-\exp n)$ and $B_n = \rho_n^{-2}$. Both radii are eventually unsaturated, and (A) implies

$$\frac{R_{\mu,\rho_n}(B_n)}{\Psi_{\mu,\rho_n}(B_n^{-1/2})} \geq c'\sqrt{1 + \log(1 + \log(1/\rho_n))} \longrightarrow \infty.$$

Thus no one finite universal C can satisfy the conjectured upper bound even at this single known law, with the algorithm knowing ρ and B. The missing statistical cost is not hidden magnitudes alone: known-profile submodels can behave differently. The lower bound uses the entire convex nuisance class to obtain a common hard pair. It does not assert that the radial variational optimizer itself is that pair, nor give a closed-form expression for the pair selected by compactness. No claim for every other magnitude law follows from (A).

A Self-contained upper checkpoint and provenance

This appendix records the existing upper argument needed for the upper inequality in Theorem 1, so the characterization does not depend on an inaccessible working-tree file. The inspected source was `research/2026-10-27-minimax-audit/theorem/critical.tex`, SHA-256

`0e8b9eaf762b3b09fa0d92ce1a036c2ceea8c12d00a5eea8286b852f36a81536`

It was read without modifying it. The following is a compact restatement of that pre-existing proof, not a new construction or discovery claim. The source is now committed at `a8a8dc4911 5aabdd1fba8d53d3390eb0a5eee864`.

Put $H = 1 + \log L$. For $j \geq 0$ choose

$$\varepsilon_j = 16^{-(j+1)}, \qquad N_j = \lceil (a^2\rho\varepsilon_j)^{-1} \rceil, \qquad K_j = \lceil \log_2(a\sqrt{N_j}) \rceil.$$

For $0 \leq k \leq K_j$, let $r_k = a2^{-(k+1)}$ and, after n observations with h_n heads, define

$$L_{k,n}^{+} = (1 + 2r_k)^{h_n}(1 - 2r_k)^{n-h_n}, \qquad L_{k,n}^{-} = (1 - 2r_k)^{h_n}(1 + 2r_k)^{n-h_n}.$$

Before N_j, stop with output zero if a negative likelihood reaches $(K_j + 1)/\varepsilon_j$, or with output one if a positive likelihood reaches $(K_j + 1)/(\rho\varepsilon_j)$. Fix zero priority for ties. At N_j, if no earlier decision was made, return the strict majority bit. Denote this bounded, observable classifier by Z_j and its work by T_j.

For the wrong-sign likelihood, each next multiplier has mean $1 - 4r_k d \leq 1$. Stopping this nonnegative supermartingale at a crossing and a deterministic horizon, and then increasing the horizon, bounds the probability of crossing level b by $1/b$. Summing over $K_j + 1$ references gives wrong-crossing probabilities at most ε_j on a positive coin and $\rho\varepsilon_j$ on a negative coin. The wrong forced-majority event is a subset of the full-tape event, of probability at most $e^{-2N_j d^2}$. This is

Hoeffding's bound: the centered Bernoulli log moment-generating function has second derivative at most $1/4$, so is at most $\lambda^2/8$; use independence and exponential Markov with $\lambda = \pm 4d$. For the exact power law,

$$\mathbb{E}_\mu e^{-2ND^2} = \frac{1 - e^{-2Na^2}}{2Na^2} \le \frac{1}{2Na^2}.$$

No independence of a stopping event and this full-tape majority event is used. It follows, uniformly over the global-rarity class, that

$$\mathbb{P}(Z_j \neq S) \le \tfrac{5}{2}\rho\varepsilon_j, \qquad \mathbb{P}(Z_j = 1, S = 0) \le \tfrac{3}{2}\rho\varepsilon_j. \tag{U1}$$

For $d \ge N_j^{-1/2}$, choose $k = \lceil \log_2(a/d) \rceil \le K_j$. Then $d/4 \le r_k \le d/2$, and the correct-sign log likelihood has mean increment at least $2d^2/3$: use

$$\tfrac{1}{2}\log(1 - 4r_k^2) + d\log\frac{1 + 2r_k}{1 - 2r_k} \ge -\tfrac{8}{3}r_k^2 + 4dr_k \ge \tfrac{2}{3}d^2.$$

Its upward increment is less than one. The bounded stopped-sum identity for the crossing time σ at a finite horizon n gives $g\mathbb{E}(\sigma \wedge n) \le A_{j,s}$, where

$$A_{j,s} = 1 + \log(K_j + 1) + \log(1/\varepsilon_j) + s\log(1/\rho).$$

Monotone convergence proves finite expected crossing time and, including the cap and the trivial small-gap case,

$$\mathbb{E}(T_j \mid d, s) \le \min\{N_j, (3/2)A_{j,s}/d^2\}. \tag{U2}$$

Write $c_0 = 3/2$. Direct integration of the negative contribution gives

$$\mathbb{E}\min\{N, c_0A/D^2\} \le (c_0A/a^2)[1 + \log_+(Na^2)].$$

For positives, layer cake and $\mathbb{E}[q\mathbf{1}_E] \le \min\{\rho, \mu(E)\}$ give

$$\mathbb{E}[q\min\{N, c_0A/D^2\}] \le \int_0^\rho \min\{N, c_0A/(a^2u)\}\,du \le (c_0A/a^2)[1 + \log_+(Na^2\rho)].$$

The exact integral is $N\rho$ when $Na^2\rho \le c_0A$, and otherwise is $(c_0A/a^2)[1 + \log(Na^2\rho/(c_0A))]$. Thus this step also enforces the global, rather than pointwise, rarity condition.

Elementary bounds from the definitions are

$$\begin{aligned} N_ja^2 &\le 2/(\rho\varepsilon_j), & K_j + 1 &\le 5(L + j + 1), \\ A_{j,0} &\le 7H(j + 1), & A_{j,1} &\le 7H(j + 1) + L, \\ 1 + \log(N_ja^2) &\le 5L(j + 1), & 1 + \log(N_ja^2\rho) &\le 5(j + 1). \end{aligned}$$

For example, $\log(L + j + 1) \le \log L + j + 1$ and $4\log 2 < 3$ prove the bound on $A_{j,0}$. Substitution in (U2) and the two integrals gives

$$\mathbb{E}T_j \le (225/2)a^{-2}LH(j + 1)^2. \tag{U3}$$

Independently choose a level J with $p_j = 2^{-(j+1)}$. For $J = 0$ return $Y = Z_0/p_0$ on one fresh coin. For $J = j \ge 1$, run Z_j and Z_{j-1} on successive unused blocks of that same coin, set $\Delta_j = Z_j - Z_{j-1}$, and return $Y = \Delta_j/p_j$. The bounded stopping times make the iid suffix argument elementary, as in the restart argument above; both decisions share the same latent sign. A disagreement implies an error of one classifier. Hence (U1) gives

$$\mathbb{E}Z_0^2 \le \tfrac{35}{32}\rho, \qquad \mathbb{E}\Delta_j^2 \le \tfrac{85}{32}\rho 16^{-j} \quad (j \ge 1).$$

Tonelli and the exact geometric sum now give

$$\mathbb{E}Y^2 \le \frac{35}{16}\rho + \frac{85}{16}\rho \sum_{j\ge 1} 8^{-j} = \frac{165}{56}\rho.$$

The disagreement bounds also give absolute summability of expected increments, so signed expectations may be telescoped after integrability is established: $\mathbb{E}Y = \lim_j \mathbb{E}Z_j = \theta$. Work costs at most twice (U3), and $\sum_j p_j(j+1)^2 = 6$, yielding

$$\mathbb{E}T_Y \le 1350a^{-2}LH.$$

All operations are on one observable coin with independent seed; the random level is unbounded but the separate expectations are finite. This completes the upper checkpoint used in Theorem 1. Its actual work also has genuinely unbounded support: at any fixed observation count n, all likelihoods are at most $(1+a)^n$. For sufficiently large j, both the smallest stopping boundary and N_j exceed the relevant finite thresholds, so $T_j > n$ on every tape. Since $\mathbb{P}(J = j) > 0$, actual work exceeds every finite cap with positive probability while retaining finite mean.

Proof of the upper bound in Theorem 1. The preceding construction supplies a one-coin unbiased output with $\sup_q \mathbb{E}_q Y^2 \le (165/56)\rho$ and $\sup_q \mathbb{E}_q T_Y \le W_* = 1350a^{-2}LH$. Independently select $N \sim \text{Poisson}(B/W_*)$ and return $\widehat{\theta} = (W_*/B)\sum_{i=1}^{N} Y_i$ from independent genuine coins. Conditioning on N proves unbiasedness, uniform expected work at most B, and variance at most $(165/56)\rho W_*/B$. All moments are finite before using these identities, by the bounds and the finite Poisson moments. Choose the zero-work constant $\rho/2$ whenever its known error bound is smaller. Both choices are observable and depend only on known parameters. With a fixed, this proves the upper inequality in (A). This established replication device is not claimed as new. □

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